Can We Teach Computer Vision to My Mother (77)? If So, How?

A one-month, feedback-led design case study and a proposed single-participant study

Non Arkaraprasertkul · Vision project | Preprint v1.0 · Author-review draft · Not peer reviewed

Abstract

Can a 77-year-old learner use a computer-vision classroom to turn domestic ambitions into realistic system requirements? This preprint treats that question as open. Vision is approximately one month old, according to its owner; its iterations respond to user feedback rather than years of collected participant data. We analyze eight retained design-audit records, four reported phone-layout comparisons, and one synthetic detector export. The selected audits document recurrent separation of pictures, instructions and actions. Recorded changes recover 101 pixels of navigation chrome and bring one teaching action 561 pixels earlier; these are interface measurements, not learning gains. Re-scoring the detector export yields no matched detections on its stylized reference images, underscoring the need to distinguish software behavior from real-world reliability. We derive an assembly-instruction teaching approach: one visible action, one captured picture, one inspectable consequence. A proposed study then evaluates teach-back, error recognition and requirement formulation for fall alerts, perimeter alerts and plant growth. No participant outcomes, age-specific effectiveness or home-safety performance are claimed.

1. The question is practical, not magical

The motivating learner is the author's mother, aged 77, as described by the project owner. She wants useful systems at home: notice a possible fall, warn about a person around the house, and track her plants. These are meaningful goals. They are not evidence that a camera can infer every event, intention or cause. The educational question is whether she can learn enough to specify what should be observed, what the system should do, and what should happen when it is wrong.

The phrase 'AI is not Harry Potter, Mom' names a mismatch between an expansive promise and a limited instrument. It should not become a rebuke. Success is not that she stops asking questions. Success is that she can ask more precise questions, challenge an answer, decide what uncertainty is acceptable, and decline a system she does not want. Her age describes the proposed participant; it does not diagnose her capabilities.

RQ1 asks which interaction barriers are documented in Vision's first month. RQ2 asks how the interface can distinguish the learner's action, the captured evidence and the machine's response. RQ3, reserved for a future study, asks whether this approach helps this learner formulate realistic, privacy-aware home-vision requirements. Only the first two can be addressed by the current artifact record.

2. Related work and the design inference

AI literacy includes understanding and critically engaging with AI, not merely operating an interface [1]. A study of 103 adults aged 50 and above reported motivation to learn alongside difficulty knowing where to begin and a preference for hands-on activity [2]. That sample is not a finding about this mother, about all people aged 77, or about Thai learners. It supports investigating a practical starting point rather than assuming one.

Human-AI interaction guidelines emphasize communicating capability and supporting correction [3]. Explanatory debugging provides a precedent for users examining and correcting a learner's behavior [4]. Teachable Machine demonstrates an approachable browser interface for making classifiers [5]. These precedents motivate keeping a user's example, the resulting prediction and the means of checking it together; they do not establish that Vision has achieved the same educational outcomes.

Multimedia-learning research treats limited processing capacity and the relationship between words and pictures as design concerns [6]. Our assembly-instruction interpretation is therefore an explicit design inference: reduce simultaneous decisions, number the sequence, put a control by the picture it changes, and show what that control actually saved. We have not experimentally compared this interpretation with IKEA instructions or with a conventional tutorial.

3. Materials and analysis

This is a retrospective artifact analysis plus a prospective protocol, not a completed learner experiment. The owner reports an approximately one-month development period and that iterations arose from user feedback. The available evidence is mediated through development records: it does not contain a complete, consented transcript corpus, a reliable count of distinct testers, recruitment details, baseline literacy scores or longitudinal participant outcomes. We therefore do not infer a participant sample size, satisfaction rate or causal educational effect.

We purposively selected eight retained audit documents dated 20 October to 3 November 2026 because they document an interaction problem, a response and verification or limitations. Table 1 lists the complete selection. The analysis script checks an evidence anchor and computes SHA-256 for every source. A single author-assigned primary theme supports an inventory, not

a reliability-tested qualitative coding scheme. A document can concern several issues; the chosen primary category is not a count of people or complaints [9].

Four before/after values were transcribed from R3 and R8. They describe English 390 by 844 pixel views of different releases. Recovered pixels equal before minus after; proportional reduction equals recovered pixels divided by before. These historical values are reported measurements in the retained audits, not fresh reruns of all historical releases. We do not pool them into an overall usability score.

A separate retained SSDLite MobileNetV2 export, generated 20 October 2026, contains predictions for the repository's evaluation split. We re-score all 100 expected frames at score threshold 0.30 and intersection-over-union 0.50 with the existing matcher. Precision is TP/(TP+FP), recall is TP/(TP+FN); absent denominators are undefined, not zero. The 200-frame reference pack separates development and evaluation by sequence. Both splits share a renderer and projection, so sequence separation does not establish independent real-world generalization.

Record / date	Documented problem to response
R1 / 10-20	Inert language controls to Shared navigation boot and readable multilingual text
R2 / 10-20	Catalogue counts competed with the visual lesson to Lesson first; catalogue details on demand
R3 / 10-23	Repeated navigation and instructions delayed the teaching action to One 44px rail and action beside picture
R4 / 10-29	Idle source labelled Studying; action readiness unclear to Instruction derived from source and example readiness
R5 / 10-30	Picture separated from source/actions by bands to Regroup instruments and remove empty gallery demand
R6 / 11-03	Need a concrete explanation of three learning methods to Actual sticker-teaching film with method comparison
R7 / 11-03	Road hidden behind an unrelated teaching exercise to Road first; optional exercise retained
R8 / 11-03	Picture, explanation and next action separated to Picture leads to adjacent explanation and action

Table 1. Complete selected corpus (R1-R8). Dates are 2026. Source paths, anchors and checksums are in the data ledger [9].

4. Findings from the available record

The single-primary-theme inventory contains five spatial-composition records, one accessibility record, one state/action record and one concept-explanation record. Five of eight does not estimate how often users encounter spatial problems: selection intentionally favored UX audits, the documents are author/agent reports, and revisions are not independent observations. The recurring design mechanism is nevertheless traceable: multiple bands or distant controls make the reader reconstruct a relationship the interface could have shown.

R3 records a reduction in closed phone chrome from 145 to 44 pixels, or 17.2% to 5.2% of an 844-pixel screen. The first teaching action moves from y=1233 to y=672. R8 records Home's picture moving from y=328 to y=117 and its next lesson link from y=872 to y=556. Table 2 gives the calculations. A smaller coordinate means earlier access in those layouts, not faster task completion, improved comprehension or a measured preference.

Accessibility and state correctness are part of the same causal chain. R1 records language controls that needed a shared navigation boot. R4 records an idle source presented as studying and readiness instructions that needed to follow actual source/example state. R6 supplies a concrete teaching film and method comparison. These records support treating a false status or an unexplained capture as an instructional defect, rather than adding more introductory prose.

The synthetic detector result is a useful negative finding. All 100 expected evaluation frames are present. At the stated thresholds there are 0 true positives, 35 false positives and 336 false negatives; aggregate precision and recall are both 0. Median recorded inference time is 35.4 ms and the 95th percentile is 57.3 ms. Those timings belong to the retained run, whose hardware is not specified in the export; they are not a fresh performance promise. Zero matches on stylized images is not a real-camera failure rate. Equally, a fast run or passing software tests cannot establish reliable fall, intrusion or plant monitoring.

Phone measure	Before	After	Gain (px)
Teach: closed top chrome	145	44	101
Teach: first teaching action y	1233	672	561
Home: picture y	328	117	211
Home: next lesson link y	872	556	316

Table 2. Historical English 390 × 844 measurements in pixels. Reductions: 69.7%, 45.5%, 64.3%, 36.2%, respectively. Coordinates are not task-completion times.

5. An assembly-instruction pedagogy

The unit of instruction is an observable cycle: predict what will happen, perform one action, inspect the exact evidence and explain the consequence. The picture dominates the workspace. An action verb names the human's task. A saved snapshot and its supplied answer remain together. The next instruction follows actual state: collect, learn, hold a new picture, check, continue. Changing a scene must be visibly different from saving it. A loading model, duplicate picture, missing source or held frame needs an honest local message.

For supervised learning, the person supplies pictures with answers. Save several visibly different examples of each class, then test an unused picture before revealing its answer. A frame from the same camera can be highly correlated with the training examples. Better same-camera performance therefore does not establish transfer to another room, view or time. In this classroom a pretrained MobileNetV2 describes the frame, and a small learner uses these descriptions; the large network is not retrained [7].

For unsupervised learning, give the machine changing pictures without answer names. It groups similar descriptions. Ask the learner what the groups seem to share, then show that lighting or viewpoint may dominate the intended concept. Finding two piles is not proof that the machine discovered the human's categories.

For reinforcement learning, distinguish an action from a label. The classroom's one-step contextual bandit chooses alert or stay quiet and receives a score for that held picture. It illustrates action feedback and exploration, but not a sequential driving policy or a home robot learning from a life history [8]. State the goal before asking for a reward. The person judges the action, not the appearance of a positive number.

A displayed vote or model confidence remains evidence to question. The classroom's nearest-neighbor vote is not a calibrated probability of correctness. Research on neural-network calibration is another reason to avoid equating an impressive percentage with certainty [10]. Show an incorrect answer, an unseen case and a case the camera cannot see. These are lesson materials, not embarrassing exceptions to hide.

Then YOU check the answer and its limits.

Figure 1. The instructional unit links a human action to captured evidence and a machine response. Seeing alone is not saving; an answer still needs checking.

6. Turn domestic ambitions into observable tasks

A home request should be rewritten as five fields: visible event, required view, response, consequence of a miss/false alert, and data policy. Table 3 applies that template to the mother's examples. The distinction between an observation and an interpretation matters: a person near a boundary is visible; criminal intent is not a visual label the system can establish from that fact alone.

For a possible fall, use drawings or safe recorded scenarios to discuss a person unexpectedly low on the floor and not moving. Do not ask an older participant to simulate a dangerous fall. Occlusion, ordinary floor activities, lighting, pets, camera failure and someone outside the view challenge the rule. An educational demo is not a validated emergency service; any real deployment needs independent testing, a human response plan and alternatives when vision fails.

For perimeter monitoring, ask which zone and hours matter, who may legitimately enter, what alert should be sent and who can review it. Evaluate missed entries and nuisance alerts without branding a person a robber or identifying them. Obtain the household's agreement about view, retention and bystanders before deciding that a camera is the right sensor.

For plant growth, first compare photographs taken from a stable viewpoint with a visible scale and known dates. Lighting, camera motion, leaf overlap and changing pose can imitate change. Define the measured quantity, such as projected leaf area or height relative to a scale; do not promise plant health, cause of disease or watering needs from that measurement alone. A ruler or a non-camera sensor may be simpler. The learner should be free to choose it.

Home goal	Five fields to agree with the learner
Possible fall	Event: unexpected low posture + inactivity. View: agreed, unobstructed room. Response: chosen check-in/helper alert. Error cost: missed help versus nuisance alarms. Data: agree processing, access and retention; avoid default recording.
Perimeter alert	Event: person in a chosen zone/time. View: agreed boundary, avoiding neighbors. Response: notify the chosen resident. Error cost: missed entry versus nuisance alerts. Data: minimize bystander capture; agree access and retention, with no identity claim.
Plant growth	Event: change relative to a scale. View: stable camera, lighting and dated observations. Response: show a comparison for the owner to inspect. Error cost: false growth from viewpoint/light. Data: agreed plant images, cropped to avoid household activity.

Table 3. Proposed five-field examples to agree with the learner. These are neither deployed systems nor validated safety or privacy guarantees.

7. Proposed study with the learner, not on her behalf

No participant sessions have been conducted for this preprint. The proposed first study is a feasibility case with one consenting participant, aged 77, who chooses her language, preferred device, text size, pace and breaks. Relationship to the author is a source of both contextual knowledge and possible deference. The invitation must make refusal and withdrawal easy, and must not make help at home conditional on participation. Obtain applicable institutional or local research review before research data collection; no approval is claimed here.

Begin with a short conversation about her three home goals and current beliefs about what a camera can see. Record her own definitions of success. Three optional sessions of roughly 20-30 minutes would cover labelled examples, grouping without names, and action feedback, each followed by a home scenario. Use stickers, ordinary objects or non-private practice footage before any live household camera. Do not retain raw household imagery by default.

At baseline, immediately after instruction and after a proposed 48-72 hour delay, ask six matched tasks using different pictures or scenarios: distinguish seeing from saving; explain a supplied label; interpret an unnamed group; score an action against a stated goal; identify an unseen or uncertain case; and formulate a bounded home request. Alternate forms reduce simple picture memorization. Ask the participant to explain, not repeat the interface wording. Record exact hints and whether an action was completed independently.

Use the rubric in Appendix A: 0 for an absent/incorrect explanation, 1 for a partially correct or content-prompted explanation, and 2 for a correct independent explanation. Preserve item-level scores and the participant's words; a 0-12 sum is descriptive and is not a validated literacy scale. Where consent permits, a second reviewer should score de-identified responses without knowing the time point. Report disagreements instead of inventing reliability from one rater.

Proposed feasibility criteria are at least four independently completed tasks out of six and an independently stated limitation for each of the three home applications. These are author-chosen planning thresholds, not established cutoffs. Report completed/attempted denominators, assistance, withdrawals, errors, delayed explanations and changes in the learner's own requests. A pre/post difference in one person cannot establish a treatment effect or generalize to older adults. If confusion persists, revise the instruction and retest rather than attributing failure to age.

8. Limitations and research integrity

The record is short, selectively preserved and produced by people and agents building the system. Feedback reports are not interchangeable with independent interviews. Primary-theme assignments have one coder. Historical layout measurements span releases and differ from current screenshots. Camera catalogue size measures listed sources, not usable views or learned knowledge. Synthetic regression fixtures do not validate household risk detection. No participant-level statistical analysis, effect size, significance test or claim of saturation is warranted.

This is a living, AI-assisted author-review manuscript. Assistance was used for literature discovery, drafting, analysis code and typesetting. Bibliographic records were checked against primary publication or author records; the author remains responsible for interpretation and approval. The analysis deliberately keeps missing learner outcomes as null rather than fabricated zeros. The paper is hosted by the project, is not peer reviewed, has no assigned DOI, and does not claim institutional endorsement.

9. Conclusion

The available evidence does not yet answer whether this mother has learned computer vision. It does show why that question should be investigated through a clearer instrument: put the human action, the captured picture and the machine's consequence in one reading path. A successful lesson should enable a person to specify an observable event, test a failure case and choose an appropriate response. It should make her a more capable co-designer, not a quieter recipient of technical promises.

Appendix A. Proposed teach-back prompts

A1 - Seeing versus saving: 'The picture is moving. What must you do to keep one picture for the lesson?' Independent success distinguishes opening/changing a view from explicitly saving one frame.

A2 - Supervised: 'Here are pictures and their names. What did you give the machine, and what should it answer for a new picture?' Independent success identifies examples plus supplied labels, and allows that a new answer may be wrong.

A3 - Unsupervised: 'It made two piles, but we gave no names. Does it know what the piles mean?' Independent success describes similarity without claiming the machine discovered the intended categories.

A4 - Reward: 'Our goal is an alert for busy traffic. It stayed quiet on this held picture. What are you scoring?' Independent success judges the action against the stated goal rather than naming the image or rewarding every attempt.

A5 - Limits: 'The camera cannot see behind this chair. No person was detected. What can we conclude?'

Independent success distinguishes no detection from no person and asks for a better view or another sensor.

A6 - Requirements: 'Choose a home task. Tell me what the camera must see, what it should do, when it may be wrong, and who can see the pictures.' Independent success provides a bounded observation, a response and a limitation/data choice. Clarifying a word is permitted; hints supplying conceptual content must be logged.

Data and reproducibility

The complete selected-record ledger, source paths, evidence anchors, SHA-256 values, all four geometry calculations and the re-scored synthetic results are published as mother77-data.json [9]. The manuscript source is mother77.json; mother77.md is the readable text export. Run node scripts/analyze-mother77.mjs, then scripts/build-mother77.py with the documented PDF runtime. These regenerate the analysis and manuscript outputs. The source record period is explicitly narrower than the owner's approximate one-month project age. No multi-year dataset is claimed. Participant outcomes remain absent until an agreed study is actually conducted.

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